Population Growth and Deforestation Emissions in Cameroon (1961–2023)

IB Environmental Systems and Societies — Standard Level · Individual Investigation

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1. Introduction and environmental context

Deforestation is one of the clearest cases of human activity altering an environmental system faster than that system can recover. Removing closed-canopy tropical forest reduces at once the carbon held in biomass, the habitat supporting high species richness, the regulation of local rainfall and the binding of soil against erosion. Forest is a store of natural capital and the services it provides are natural income. Clearing it replaces a largely self-regulating system, held near a steady state by internal negative feedback, with a simplified agricultural system that stays productive only through continuous human input. Recovery is slow: regrowth may restore canopy within decades but not the species composition or soil structure of primary forest.

Cameroon lies at the north-western edge of the Congo Basin, the world's second largest block of tropical rainforest. Clearing forest releases the carbon stored in trees and soil, so deforestation appears in national greenhouse-gas accounts as CO_2 emissions from land-use change. For Cameroon this flow is not a minor category: in 2009 it reached 81.9 Mt CO_2, and even in 2023 it was 23.1 Mt, roughly 2.4 times the country's fossil-fuel emissions of 9.7 Mt. Cameroon's climate impact is therefore mainly a land-use story rather than an energy one, which is typical of forested low-income countries and makes land-use-change emissions a useful single measure of how heavily forests are being converted.

What drives that conversion is contested. Tyukavina et al. (2018) found that forest disturbance across the Congo Basin between 2000 and 2014 was dominated by small-scale, largely manual clearing for agriculture rather than by industrial logging, and that the annual rate of this clearing almost doubled. Because smallholder clearing is carried out by households, the common explanation is population pressure: more people require more farmland, and farmland comes from forest. Cocoa, the region's leading export tree crop, is often named as the driver — Ordway et al. (2017) identified cocoa as the fastest-expanding export crop in sub-Saharan Africa, with most expansion occurring outside industrial plantations. The population-pressure argument is intuitive and widely repeated, but it is an empirical claim, and it can be tested against national data rather than assumed.

2. Research question and rationale

To what extent is population growth associated with CO_2 emissions from land-use change in Cameroon between 1961 and 2023?

The question matters because the answer shapes policy. If forest loss tracks population, the response is to raise agricultural yields so that a growing population needs less new land. If it does not, attention belongs instead with export markets, roads, logging and governance, and blaming population growth misdirects effort onto the households with the least capacity to change.

Independent variable: the annual increase in Cameroon's population (people per year). **Dependent variable:** annual CO_2 emissions from land-use change in Cameroon (million tonnes, Mt CO_2 per year). **Controlled aspects:** one country, one dataset version, identical calendar years for both variables, and the same statistical test throughout.

Population is used as a flow rather than a level for a specific reason. Total population is a *stock* — the number of people alive in a given year — whereas land-use-change emissions are a *flow*, the carbon released during that year. Correlating a stock against a flow across six decades would mostly measure whether two series happen to trend together over time. Defining population growth as the **annual increase** puts both variables on a per-year basis, which is the defensible comparison. The levels correlation is still reported, but as a check on that reasoning rather than as the main result.

Boundaries. Cameroon is the spatial unit because both variables are reported to consistent national boundaries. The period begins in 1961, the first year after independence and the start of continuous modern statistics, and ends in 2023, the last complete year in the emissions dataset.

3. Strategy and perspectives

The issue sits inside a genuine conflict, and the groups involved hold different environmental value systems that lead them to read the same forest differently.

Smallholder farmers. For rural households, forest is a land reserve and cash crops are one of the few reliable sources of income. Clearing a hectare is rational for a household with limited capital: forest soils are fertile, the land is often effectively free to occupy, and a cocoa or oil palm plot yields income for decades. This anthropocentric position is not an uninformed one — it reflects real constraints rather than indifference to forests.

The national economy. Agricultural exports earn foreign exchange and employ a large share of the workforce, so the state has an interest in sustained output. This creates a tension within government, since the same state holds forest-protection commitments. Restricting expansion carries a measurable economic cost, which is why forest policy is often weakly enforced.

Conservation interests. From an ecocentric standpoint, Congo Basin forest has value as habitat, as a carbon store and as a regulator of regional hydrology — value largely non-monetised and accruing globally rather than locally. That asymmetry is central: the benefits of clearing are captured locally and immediately, while the costs of lost ecosystem services are global and deferred. This externality explains why the issue is not resolved by farmers simply choosing better.

Where responsibility is placed. A population-pressure framing is itself a perspective with consequences. It locates the cause in the fertility of poor rural households rather than in consumption elsewhere, and it has historically been used to justify coercive policy. Testing the claim rather than assuming it is therefore an ethical question as well as a scientific one.

Three strategies are advanced, each with real drawbacks.

Agroforestry. Growing cocoa under shade retains part of the canopy, supports more biodiversity than full-sun monoculture, stores more carbon and protects soil. It is already the dominant system in Cameroon. Its weakness is yield: shaded cocoa produces less per hectare, so if demand is fixed, lower yields require *more* land. This is the land-sharing versus land-sparing tension. Gockowski and Sonwa (2011) modelled the alternative and estimated that if intensified cocoa technology had been adopted from the 1960s across Côte d'Ivoire, Ghana, Nigeria and Cameroon, more than 21,000 km^2 of deforestation and degradation and roughly 1.4 billion tonnes of CO_2 emissions could have been avoided. Their conclusion is uncomfortable for conservation thinking: intensification, though it reduces on-farm biodiversity, may spare more forest overall. Which strategy is "better" depends on whether value is placed on biodiversity within the farm or on total forest left standing.

Certification. Schemes such as Rainforest Alliance and Fairtrade reward deforestation-free production with price premiums. They give consumers leverage and can transfer income, but depend on audit quality, cover only part of production, and are vulnerable to leakage: if certified farms stop expanding while demand continues, clearing moves elsewhere.

Regulation. The EU Deforestation Regulation (European Union, 2023) requires importers to show that commodities including cocoa were not grown on land deforested after 2020. It is the

strongest instrument available, but places the compliance burden on smallholders least able to carry it, raising a question of equity: who pays to protect a global public good?

These perspectives shape outcomes directly: because the most economically rational household choice removes forest, and the resulting costs fall largely outside Cameroon, forest loss continues even where all parties agree it is undesirable.

4. Methodology

This investigation uses **exclusively secondary quantitative data**: no fieldwork, interviews, questionnaires or experiments were involved. It is a reproducible desk-based correlational analysis.

Source. All variables come from the CO_2 and Greenhouse Gas Emissions dataset published by Our World in Data, which distributes a single compiled file through its public repository. The underlying sources are the Global Carbon Budget (Friedlingstein et al., 2025) for emissions and the Maddison Project Database (Bolt & van Zanden, 2025) for GDP.

Variable used	Column	Unit	Original source	Coverage obtained
CO_2 from land-use change	`land_use_change_co2`	million tonnes (Mt) yr^{-1}	Global Carbon Budget (2025)	1961–2023, complete
Population	`population`	people	Our World in Data population series (2024)	1961–2023, complete
GDP	`gdp`	international -$, 2011 prices	Maddison Project Database 2023	1961–2022 (2023 missing)
Fossil CO_2 (context)	`co2`	million tonnes (Mt) yr^{-1}	Global Carbon Budget (2025)	1961–2023, complete

Selection. An agricultural land-area variable would have matched the research question more directly, so World Bank *agricultural land* and *permanent cropland* series for Cameroon were examined first and **rejected**. In those series 37% and 46% of years respectively showed exactly zero change from the previous year, and permanent cropland was identical in all eight years from 2011 to 2018. Values repeated to that degree are carried forward rather than measured, and correlating them would produce a coefficient describing a reporting convention rather than the environment. Land-use-change CO_2 was used instead: it has 62 usable years with no repeated consecutive values.

Extraction and processing. The compiled file was downloaded, filtered to Cameroon and to 1961–2023, and the four columns above were retained. Annual population increase was computed as $\Delta P_t = P_t - P_{t-1}$, which necessarily loses the first year. Land-use-change emissions per person were computed to separate total from per-head intensity. All steps are scripted, so every figure in this report traces back to the unmodified download.

Statistical treatment. Spearman's rank correlation (r_s) is the primary test. Shapiro–Wilk tests reject normality for both land-use-change emissions (p = < 0.001) and annual population increase (p = < 0.001), so Pearson's assumption of bivariate normality does not hold; Spearman requires only a monotonic relationship and resists outliers. Pearson's r is reported alongside for comparison. For every coefficient the sample size, two-tailed p-value and the critical value at $\alpha = 0.05$ are given.

Four further tests address specific threats to validity:

1. **Levels comparison** — total population against emissions, to show how much apparent association comes from shared time trend.
2. **First differences** — both variables differenced, removing any common trend.
3. **Lagged correlation (0–6 years)** — clearing may precede the emissions accounting, and new farmland may be cleared in anticipation of population growth.
4. **Sub-period split at 1990** — to test whether one relationship holds across six decades of very different economic conditions.

5. Data presentation and processing

The full annual dataset is given in Appendix A and summarised below. Table 3 reports every correlation computed, so that no test is hidden; reporting only the strongest would misrepresent the evidence.

Table 1. Cameroon, 1961–2023: population, annual population increase, CO_2 from land-use change, land-use-change CO_2 per person, fossil CO_2 and GDP per capita. *(Full table in Appendix A; source: Global Carbon Budget 2025 and Maddison Project Database 2023, via Our World in Data.)*

Table 2. Decade means and distribution statistics, with Shapiro–Wilk normality results.

Table 3. All correlation results: levels, annual change, first differences, GDP per capita, lags 0–6 and both sub-periods, each with n, r_s, r, two-tailed p and the critical value.

Figure 1. Population and land-use-change emissions, 1961–2023, plotted as two panels on separate scales rather than on a shared dual axis, which would allow the apparent relationship to be altered simply by rescaling. **Figure 2.** Scatter of land-use-change emissions against annual population increase, with least-squares line and coefficients. **Figure 3.** Land-use-change emissions per person, 1961–2023. **Figure 4.** The same scatter split into 1961–1990 and 1991–2023. **Figure 5.** Spearman's r_s against lag (0–6 years), with the $\alpha = 0.05$ threshold marked. **Figure 6.** Cameroon's per-person land-use-change emissions against five regional neighbours.

6. Analysis and discussion

Trends. Between 1961 and 2023 Cameroon's population rose from 5.24 million to 28.37 million, a factor of 5.4, increasing in every single year. Land-use-change emissions did not follow. They moved from 44.7 Mt to 23.1 Mt, a change of -48%, fluctuating around a mean of 33.5 Mt with a standard deviation of 11.6 Mt and no steady direction: decade means fell from 43.6 Mt in the 1970s to 23.7 Mt in the 1990s, then rose again to 35.0 Mt in the 2000s. One variable grows monotonically; the other oscillates. That alone makes a strong positive association unlikely.

Primary result. Annual population increase against land-use-change emissions gives r_s = -0.354 (p = 0.005, n = 62, critical |r| = 0.250) — a weak and statistically significant association, but **negative**. Years of faster population growth were, if anything, years of *lower* deforestation emissions. The levels comparison agrees (r_s = -0.386, p = 0.002). Pearson's r = -0.184 (p = 0.152) is much weaker and not significant. The two tests disagreeing is informative rather than awkward: it indicates a relationship that is monotonic in rank but not linear, with extreme years — including an outlier of 81.9 Mt in 2009 — pulling the least-squares line while barely affecting the ranks. Reporting Pearson alone would have concealed the pattern; reporting it alone as significant would have overstated it.

The association does not survive detrending. With both variables expressed as year-on-year changes, r_s falls to -0.006 (p = 0.964, n = 62) — not statistically significant, and essentially zero. This is the single most important result. It shows that the negative coefficient is a property of the two long-run trajectories, not of any year-to-year link: population rose steadily while emissions drifted down, so ranking them against each other produces a negative coefficient even though a year of unusually fast population growth tells us nothing about that year's emissions. The lag tests point the same way, with the strongest coefficient at lag 0 (-0.354) and no peak at any plausible delay.

The relationship reverses between sub-periods. For 1961–1990, r_s = -0.444 (p = 0.016, n = 29); for 1991–2023 it is 0.324 (p = 0.066, n = 33). The sign flips. A single coefficient for the whole period therefore averages two opposite patterns, and the headline answer depends on where the period boundaries are drawn — a reminder that a correlation computed over an arbitrary span can be an artefact of that span.

Decoupling. The clearest finding is in intensity. Land-use-change emissions per person fell from 8.54 t CO_2 in 1961 to 0.81 t in 2023, a decline of 90%. Cameroon added roughly 5.4 times its 1961 population while the carbon released per head from land conversion fell by an order of magnitude. This is relative decoupling: forest is still being converted, but far less per additional person. Figure 6 shows the same downward path across the region, so this is not unique to Cameroon.

Why population growth may not drive clearing proportionally. Several mechanisms are consistent with the data. Much population growth has been absorbed by cities rather than by new farms, so additional people do not translate into proportionally more cleared land. Yields on existing land have risen. The most accessible forest near roads and settlements was converted early, so later clearing faces higher costs. Economic conditions matter independently of population: the collapse in commodity prices and the structural adjustment period of the late 1980s and 1990s coincide with the lowest emissions decade in the series, which suggests that the

ability to invest in new plantations constrains clearing more than the number of people does. Consistent with this, GDP per capita also correlates negatively with emissions (r_s = -0.314, p = 0.013), so affluence is not substituting for population as an explanation either.

Alternative explanations and limits. Land-use-change emissions aggregate every cause of carbon loss from land: smallholder clearing, logging, oil palm, fuelwood, fire and peat, minus regrowth uptake. A fall could reflect less clearing, or clearing of land holding less carbon, or stronger regrowth. These cannot be separated here. Crucially, this variable is **modelled rather than measured**, so part of its year-to-year movement reflects model assumptions. For all these reasons **no causal claim is made**: the analysis shows that population growth and deforestation emissions in Cameroon have not moved together, not that population growth has no effect on forests.

7. Conclusion

Answering the research question directly: population growth is associated with CO_2 emissions from land-use change in Cameroon between 1961 and 2023 only to a weak extent, and in the **opposite direction** to the population-pressure hypothesis. The primary test gives r_s = -0.354 (p = 0.005, n = 62), which is statistically significant and negative. That association does not survive detrending — first differences give r_s = -0.006 (p = 0.964), not statistically significant — and it reverses sign between the two sub-periods. The defensible conclusion is that at national scale there is **no stable year-to-year relationship** between how fast Cameroon's population grew and how much carbon its land-use change released, while emissions per person fell by 90%.

The objectives were largely achieved: two complete annual series were assembled, tested with appropriate non-parametric statistics, and checked against three separate threats to validity. The investigation did not, and at national resolution could not, identify what does drive Cameroon's deforestation. Every statement here rests on the data presented.

8. Evaluation

Reliability of the data. The Global Carbon Budget is peer-reviewed, revised annually and transparently documented, and population coverage is complete for every year. The serious qualification is that land-use-change emissions are **output of bookkeeping models**, not measurements. They combine land-use statistics with assumed carbon densities, and this term carries the largest uncertainty in the entire global carbon budget. Treating these figures as observations would overstate their precision, and because estimates are revised when the models change, a later release could alter the correlations reported here. GDP is also missing for 2023.

Validity. Land-use-change CO_2 is a proxy for deforestation, not a measure of it. It is a *net* flux that includes degradation and peat and subtracts regrowth uptake, and it depends on the carbon density of the land cleared. A fall in emissions is therefore consistent with less clearing, with clearing of lower-carbon land, or with greater regrowth — three environmentally different outcomes that the data cannot distinguish. The independent variable is also crude: **total** population counts urban residents, whose land demand differs completely from that of farming households. Rural population would have been the better variable and is the clearest single improvement available.

Limitations of national data. Aggregating to the national total averages the forested south, where clearing occurs, with the drier north, where little forest remains. Inferring household behaviour from national aggregates risks the ecological fallacy: a national correlation does not describe what individual families do.

Statistical limitations. With $n = 62$, a coefficient must exceed 0.250 to reach significance, so only a fairly strong relationship is detectable. More importantly, thirteen correlations were computed; at $\alpha = 0.05$ roughly one in twenty would be expected to appear significant by chance, so marginal results such as the 1991–2023 sub-period ($p = 0.066$) should not be read as findings. Correlation cannot establish causal direction or exclude a third variable. The sign reversal between sub-periods shows directly that the whole-period coefficient is unstable.

Confounding variables not controlled include urbanisation rate, world commodity prices, structural adjustment and debt, oil revenue, timber concession activity, road construction, fuelwood demand and conflict-driven displacement — several of which plausibly affect both variables.

Improvements. Substituting rural for total population would sharpen the test immediately. Repeating the analysis with a directly observed measure of forest area — FAO Forest Resources Assessment figures or satellite tree-cover-loss data — would remove dependence on modelled emissions. Working at regional rather than national scale would remove most of the aggregation problem. A multiple regression including population, income and commodity prices would begin to separate drivers that correlation cannot.

Further research. The attribution question is better answered by spatial overlay than by national correlation: combining district-level population with satellite forest-loss maps would show directly whether clearing happens where people are increasing. Comparing Cameroon with Côte d'Ivoire and Ghana, where cocoa is mainly full-sun and forest loss has been heavier, would test whether shaded cultivation genuinely reduces forest loss per tonne produced.

9. References

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10. Appendices

Appendix A — Complete dataset. All 63 annual observations (`output/table1_cameroon.csv`): year, population, annual population increase, land-use-change CO_2, land-use-change CO_2 per person, fossil CO_2 and GDP per capita, exactly as extracted, with source and access date.

Appendix B — Statistical calculations. Worked Spearman calculation for the primary test (ranks, rank differences, Σd^2, substitution into $r_s = 1 - 6\Sigma d^2/[n(n^2-1)]$), the tie-corrected value, the full correlation table, Shapiro–Wilk output and critical values at $\alpha = 0.05$.

Appendix C — Extraction and processing record. Source URL, file size, date of access, the columns retained, and the complete analysis script, so the procedure can be repeated exactly.

Appendix D — Rejected datasets. World Bank agricultural-land and permanent-cropland series for Cameroon, with the counts of zero-change years that led to their rejection, recorded so that the selection decision is transparent and checkable.

Appendix E — Supplementary figures. Decade-mean emissions; the regional comparison (Figure 6); sub-period scatter plots shown separately.

Word count note

The countable total excludes, per IB requirements: title page, contents, section headings, tables and their captions, figures and their captions, equations and calculations, in-text citations, the reference list and the appendices. `scripts/wordcount.py` reproduces the figure.